Storing Data

Storage : The purpose of a storage device is to hold data—even when the computer is turned off—so the data can be used whenever it is needed. Storage involves two processes:

- Writing, or recording, the data so it can be found later for use.
- Reading the stored data, then transferring it into the computer's memory.

The physical materials on which data is stored are called storage media. The hardware components that write data to, and read data from, storage media are called storage devices.

For example, a diskette is a storage medium (medium is the singular form of the word media); a diskette drive is a storage device.

- **Categorizing Storage Devices:**

The two main categories of storage technology used today

- ❑ **Magnetic Storage.**
- ❑ **Optical Storage.**

A third category of storage—solid-state storage—is increasingly being used in computer systems, but is more commonly found in devices such as digital cameras and media players.

Magnetic Storage Devices

- Most common form of storage.
- Hard drives, floppy drives tape.
- All magnetic drives work the same.
- diskette drives, hard disk drives, high-capacity floppy disk drives, and rape drives use similar techniques for writing and reading data.
- The surfaces of diskettes, hard disks, high-capacity floppy disks, and magnetic tape are coated with a magnetically sensitive material, such as iron oxide, that reacts to a magnetic field.

Floppy Disk

Hard Disk

Tape

- Data storage and retrieval

✔ Media is covered with iron oxide.

✔ Read/Write head is a magnet.

✔ Magnet writes charges on the media.

- Positive charge is a 1
- Negative charge is 0

✔ Magnet reads charges

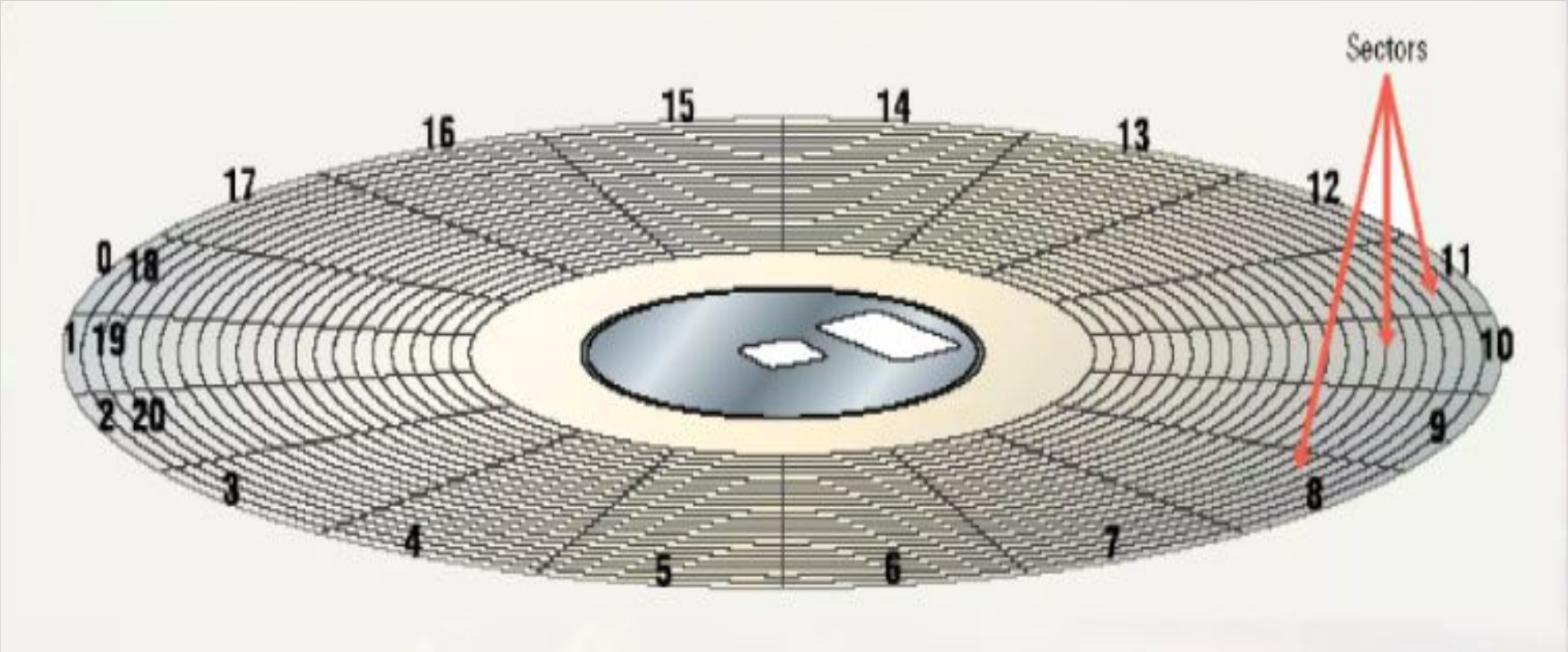
✔ Drive converts charges into binary

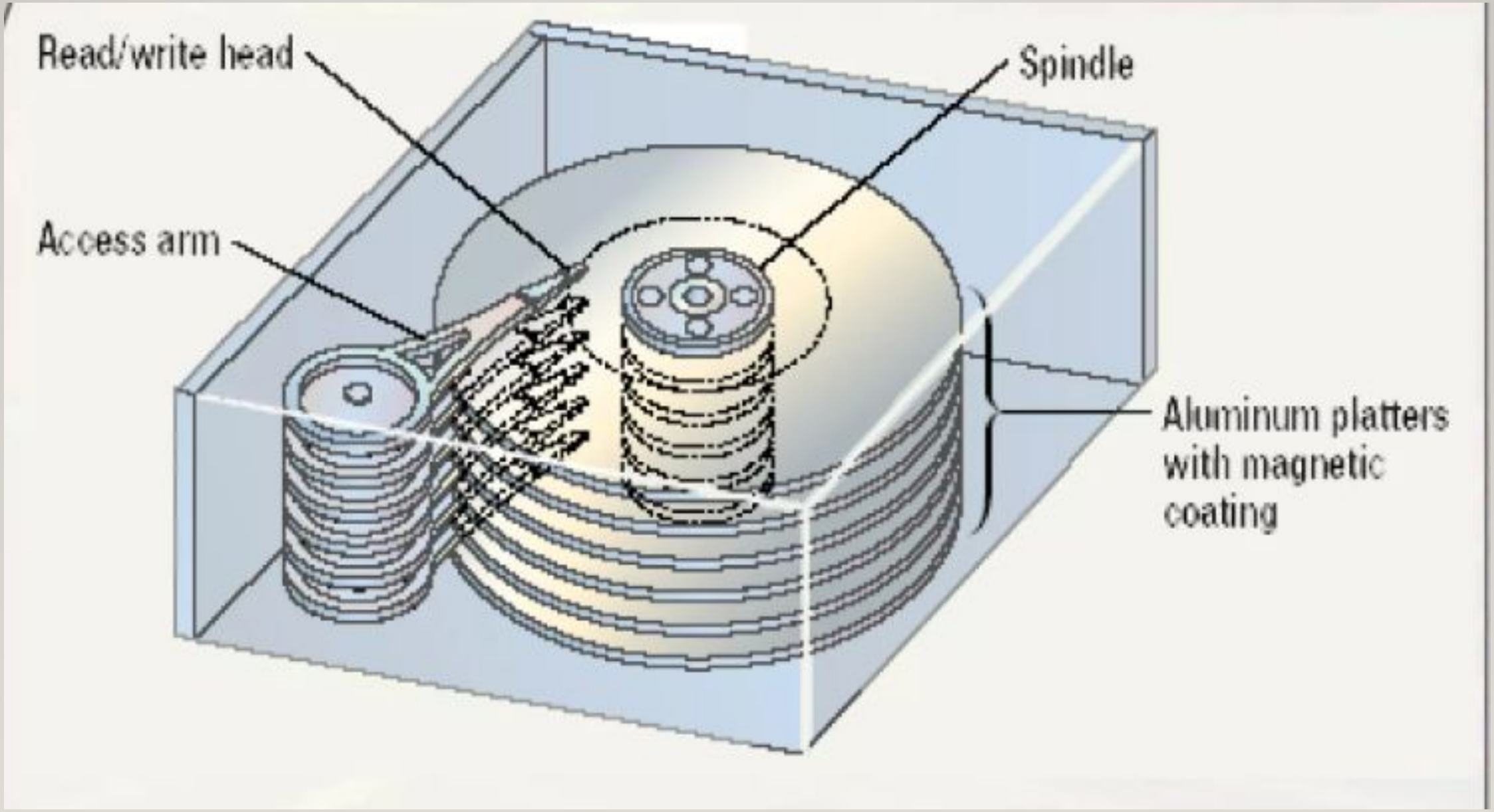
✔ Magnetic storage devices work by polarizing tiny pieces of iron on the magnetic medium. Read/write heads contain electromagnets that create magnetic charges on the medium.

Data Organization

- Disks must be formatted before use.
- Format draws tracks on the Disk
- Tracks is divided into sectors
- Amount a data drive can read.

Tracks and Sectors




Read/write head
Spindle
Access arm
Aluminum platters
with magnetic
coating

MAGNETIC

PROS	CONS
• Available with **large capacities** and **cheap to buy**	• Generally not **durable** (disks can break easily)
• Usually **reliable**, although failures do occur	• **Slower read/write** times compared to SSDs

Optical Storage Devices:

- Thee most popular alternatives to magnetic storage systems are optical systems, including CD-ROM, DVD-ROM, and their variants.
- These devices fall into the category of optical storage because they store data on a reflective surface so it can be read by a beam of laser light.

DVD-ROM

- Digital Video Disk
- Use both sides of the disk
- Capacities can reach 18 GB
- DVD players can read CDs

OPTICAL

~700 MB ~5 GB ~25 GB

- Examples include CDs, DVDs, and Blu-Ray

- They use different <u>reflections of light</u> to **read** data, and use a <u>laser</u> to **write** data

- The information is held on the disc, but you need a **disc drive** to access it

CD Recordable (CD-R)

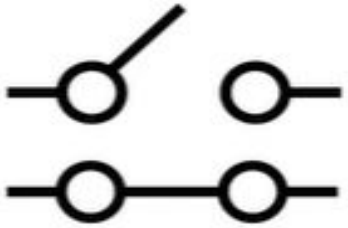
- Create a data or audio CD
- Data cannot be changed
- Can continue adding until full

PROS	CONS
• **Reliable**	• Per unit, have a smaller **capacity** than alternative
• **Good portability** – easy to pack and carry	• **Slower read/write speed** compared to alternatives
	• Need a **disc drive** to use

Solid-state storage:

- Solid-state storage devices store data on memory circuits rather than disks or tapes. They store data electronically, not in a magnetic or optical form.
- Solid-state storage devices can use either volatile or nonvolatile memory: Examples of such devices include flash memory. smart cards, and solid-state disks.

SOLID STATE



- Have no moving parts
- Electronic circuits are used to store the data
- Examples include SSDs and **flash memory** devices like USB memory sticks and memory cards (like SD cards)

SOLID STATE

PROS	CONS
• Generally **reliable**	• **Higher cost** per GB
• Can be very **durable**	• Sometimes have a **limited number of writes** (leading to reliability problems long term)
• Can be **portable**	
• Generally **faster read/write** times	